

● HARD

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

07

10 answers · Rationale-rich · Teach from doc

Q1**A****A.** $-18x + 280^\circ$

Choice A is correct. It's given that two lines intersect at exactly one point, forming two acute angles and two obtuse angles. When two lines intersect, opposite angles are congruent and adjacent angles are supplementary. It's given that the measure of one of the angles is $(9x - 140)^\circ$, so the measure of the opposite angle is $(9x - 140)^\circ$, and the measure of each adjacent angle is $(180 - (9x - 140))^\circ$, or $(-9x + 320)^\circ$. The possible sums of the measures of any two of the angles are $(9x - 140)^\circ + (9x - 140)^\circ$, or $(18x - 280)^\circ$, $(-9x + 320)^\circ + (-9x + 320)^\circ$, or $(-18x + 640)^\circ$, and $(9x - 140)^\circ + (-9x + 320)^\circ$, or 180° . Of the given choices, only $(-18x + 280)^\circ$ is not equivalent to any of these expressions and could not be the sum of the measures of any two of the angles.

Choice B is incorrect. This is the sum of the measures of two angles that are adjacent to the given angle.

Choice C is incorrect. This is the sum of the measure of the given angle and the measure of its opposite angle.

Choice D is incorrect. This is the sum of the measure of the given angle and the measure of an angle adjacent to the given angle.

Q2**C****C.** 20

Choice C is correct. It's given that parallelogram $ABCD$ is similar to parallelogram $PQRS$. When two parallelograms are similar, if the scale factor between their corresponding side lengths is k , the scale factor between their areas is k^2 . It's given that the length of each side of parallelogram $PQRS$ is 2 times the length of its corresponding side of parallelogram $ABCD$, so the scale factor between their corresponding side lengths is 2. It follows that the scale factor between their areas is 2^2 , or 4. It's given that the area, in square centimeters, of parallelogram $ABCD$ is 5. It follows that the area, in square centimeters, of parallelogram $PQRS$ is $5(4)$, or 20.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**-52**

The correct answer is -52. The equation of a circle in the xy -plane with its center at h, k and a radius of r can be written in the form $x - h^2 + y - k^2 = r^2$. It's given that a circle in the xy -plane has its center at $-5, 2$ and has a radius of 9. Substituting -5 for h , 2 for k , and 9 for r in the equation $x - h^2 + y - k^2 = r^2$ yields $x - 5^2 + y - 2^2 = 9^2$, or $x + 5^2 + y - 2^2 = 81$. It's also given that an equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants.

Therefore, $x + 5^2 + y - 2^2 = 81$ can be rewritten in the form $x^2 + y^2 + ax + by + c = 0$. The equation $x + 5^2 + y - 2^2 = 81$, or $x + 5x + 5 + y - 2y - 2 = 81$, can be rewritten as $x^2 + 5x + 5x + 25 + y^2 - 2y - 2y + 4 = 81$. Combining like terms on the left-hand side of this equation yields $x^2 + y^2 + 10x - 4y + 29 = 81$. Subtracting 81 from both sides of this equation yields $x^2 + y^2 + 10x - 4y - 52 = 0$, which is equivalent to $x^2 + y^2 + 10x + -4y + -52 = 0$. This equation is in the form $x^2 + y^2 + ax + by + c = 0$. Therefore, the value of c is -52.

Q4**.8823**

Accept also: .8824, 15/17

The correct answer is $\frac{15}{17}$. It's given that angle J is the right angle in triangle JKL . Therefore, the acute angles of triangle JKL are angle K and angle L . The hypotenuse of a right triangle is the side opposite its right angle. Therefore, the hypotenuse of triangle JKL is side KL . The cosine of an acute angle in a right triangle is the ratio of the length of the side adjacent to the angle to the length of the hypotenuse. It's given that $\cos K = \frac{24}{51}$. This can be written as $\cos K = \frac{8}{17}$. Since the cosine of angle K is a ratio, it follows that the length of the side adjacent to angle K is $8n$ and the length of the hypotenuse is $17n$, where n is a constant. Therefore, $JK = 8n$ and $KL = 17n$. The Pythagorean theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides. For triangle JKL , it follows that $JK^2 + JL^2 = KL^2$. Substituting $8n$ for JK and $17n$ for KL yields $8n^2 + JL^2 = 17n^2$. This is equivalent to $64n^2 + JL^2 = 289n^2$. Subtracting $64n^2$ from each side of this equation yields $JL^2 = 225n^2$. Taking the square root of each side of this equation yields $JL = 15n$. Since $\cos L = \frac{JL}{KL}$, it follows that $\cos L = \frac{15n}{17n}$, which can be rewritten as $\cos L = \frac{15}{17}$. Note that $15/17$, $.8824$, $.8823$, and 0.882 are examples of ways to enter a correct answer.

Q5**C**C. $90 - t^\circ$

Choice C is correct. Since $P = 4, 5$ and $Q = 4, 7$, side PQ is parallel to the y -axis and has a length of 2. Since $P = 4, 5$ and $R = 6, 5$, side PR is parallel to the x -axis and has a length of 2. Therefore, triangle PQR is a right isosceles triangle, where $\angle P$ has measure 90° and $\angle Q$ and $\angle R$ each have measure 45° . It follows that if the measure of $\angle Q$ is t° , then $t = 45$. Since $L = 4, 5$ and $M = 4, 7 + k$, side LM is parallel to the y -axis and has a length of $k + 2$. Since $L = 4, 5$ and $N = 6 + k, 5$, side LN is parallel to the x -axis and has a length of $k + 2$. Therefore, triangle LMN is a right isosceles triangle, where $\angle L$ has measure 90° and $\angle M$ and $\angle N$ each have measure 45° . Of the given choices, only $90 - t^\circ$ is equal to 45° , so the measure of $\angle N$ is $(90 - t)^\circ$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**A**

A. $\frac{1}{3}$

Choice A is correct. The equation for the volume of a right circular cone is $V = \frac{1}{3} \pi r^2 h$. It's given that the volume of the right circular cone is $\frac{1}{3} \pi$ cubic feet and the height is 9 feet.

Substituting these values for V and h, respectively, gives $\frac{1}{3} \pi = \frac{1}{3} \pi r^2 (9)$. Dividing both sides of the equation by $\frac{1}{3} \pi$ gives $1 = r^2 (9)$. Dividing both sides of the equation by 9 gives $\frac{1}{9} = r^2$.

Taking the square root of both sides results in two possible values for the radius, $\sqrt{\left(\frac{1}{9}\right)}$ or $-\sqrt{\left(\frac{1}{9}\right)}$. Since the radius can't have a negative value, that leaves $\sqrt{\left(\frac{1}{9}\right)}$ as the only

possibility. Applying the quotient property of square roots, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$, results in $r = \frac{\sqrt{1}}{\sqrt{9}}$, or $r = \frac{1}{3}$.

Choices B and C are incorrect and may result from incorrectly evaluating $\sqrt{\left(\frac{1}{9}\right)}$. Choice D is incorrect and may result from solving $r^2 = 9$ instead of $r^2 = \frac{1}{9}$.

Q7**C**

C. $x - 11^2 + y + 3^2 = 1$

Choice C is correct. The equation of a circle in the xy -plane can be written as

$(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r units. It's given that circle A has the equation $(x - 7)^2 + (y + 3)^2 = 1$, which can be written as $(x - 7)^2 + (y - (-3))^2 = 1^2$. It follows that $h = 7$, $k = -3$, and $r = 1$. Therefore, the center of circle A is $(7, -3)$ and its radius is 1 unit. If circle A is translated 4 units to the right, the x -coordinate of the center will increase by 4, while the y -coordinate and the radius of the circle will remain unchanged. Translating the center of circle A to the right 4 units yields $(7 + 4, -3)$, or $(11, -3)$. Therefore, the center of circle B is $(11, -3)$. Substituting 11 for h , -3 for k , and 1 for r into the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 11)^2 + (y - (-3))^2 = 1$, or $(x - 11)^2 + (y + 3)^2 = 1$. Therefore, the equation $(x - 11)^2 + (y + 3)^2 = 1$ represents circle B.

Choice A is incorrect. This equation represents a circle obtained by shifting circle A down, rather than right, 4 units.

Choice B is incorrect. This equation represents a circle obtained by shifting circle A left, rather than right, 4 units.

Choice D is incorrect. This equation represents a circle obtained by shifting circle A up, rather than right, 4 units.

Q8**B****B.** $26\sqrt{37}$

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that in triangle ABC , angle B is a right angle. Therefore, triangle ABC is a right triangle, where the hypotenuse is side AC and the legs are sides AB and BC . It's given that the lengths of sides AB and BC are $10\sqrt{37}$ and $24\sqrt{37}$, respectively. Substituting these values for a and b in the formula $c^2 = a^2 + b^2$ yields $c^2 = 10\sqrt{37}^2 + 24\sqrt{37}^2$, which is equivalent to $c^2 = 10037 + 57637$, or $c^2 = 67637$. Taking the square root of both sides of this equation yields $c = \pm 26\sqrt{37}$. Since c represents the length of the hypotenuse, side AC , c must be positive. Therefore, the length of side AC is $26\sqrt{37}$.

Choice A is incorrect. This is the result of solving the equation $c = 24\sqrt{37} - 10\sqrt{37}$, not $c^2 = 10\sqrt{37}^2 + 24\sqrt{37}^2$.

Choice C is incorrect. This is the result of solving the equation $c = 10\sqrt{37} + 24\sqrt{37}$, not $c^2 = 10\sqrt{37}^2 + 24\sqrt{37}^2$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**either**

The correct answer is either $\frac{10}{3}$, $\frac{15}{4}$, or $\frac{25}{6}$. The area of triangle ABC can be expressed as $\frac{1}{2}(5+7)y$ or $6y$. It's given that the area of triangle ABC is at least 48 but no more than 60. It follows that $48 \leq 6y \leq 60$. Dividing by 6 to isolate y in this compound inequality yields $8 \leq y \leq 10$. Since y is an integer, $y = 8, 9, \text{ or } 10$. In the given figure, the two right triangles shown are similar because they have two pairs of congruent angles: their respective right angles and angle A. Therefore, the following proportion is true: $\frac{x}{y} = \frac{5}{12}$. Substituting 8 for y in the proportion results in $\frac{x}{8} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{10}{3}$. Substituting 9 for y in the proportion results in $\frac{x}{9} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{15}{4}$. Substituting 10 for y in the proportion results in $\frac{x}{10} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{25}{6}$. Note that 10/3, 15/4, 25/6, 3.333, 3.75, 4.166, and 4.167 are examples of ways to enter a correct answer.

Q10

D

D. 1,944

Choice D is correct. The area of a rectangle is given by bh , where b is the length of the base of the rectangle and h is its height. Let x represent the length, in units, of the base of rectangle ABCD, and let y represent its height, in units. Substituting x for b and y for h in the formula bh yields xy . Therefore, the area, in square units, of ABCD can be represented by the expression xy . It's given that the length of each side of EFGH is 6 times the length of the corresponding side of ABCD. Therefore, the length, in units, of the base of EFGH can be represented by the expression $6x$, and its height, in units, can be represented by the expression $6y$. Substituting $6x$ for b and $6y$ for h in the formula bh yields $6x6y$, which is equivalent to $36xy$. Therefore, the area, in square units, of EFGH can be represented by the expression $36xy$. It's given that the area of ABCD is 54 square units. Since xy represents the area, in square units, of ABCD, substituting 54 for xy in the expression $36xy$ yields 3654, or 1,944. Therefore, the area, in square units, of EFGH is 1,944.

Choice A is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{1}{6}}$, not 6, times the length of the corresponding side of ABCD.

Choice B is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{2}{3}}$, not 6, times the length of the corresponding side of ABCD.

Choice C is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{6}$, not 6, times the length of the corresponding side of ABCD.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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